

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is uniformly text-only MSA [Q2, Q92], aligning with the deployment's text-only modality [A4] and MSA target [elicitation]. Multimodal content was discarded [Q37] and contextual passages preserved where needed [Q38]. However, the benchmark's MSA is fluent native-speaker register from school exams, while the deployment's users are non-native learners with basic-to-intermediate MSA proficiency, likely producing grammatically imperfect or code-switched Arabic [elicitation, expected_arabic_proficiency]. This signal-distribution gap is structural and undocumented as a design trade-off in the paper. The benchmark also notes that real-world Arabic LLM exposure includes dialectal Arabic, which is excluded here [Q92]. One minor data-quality issue observed: an Arabic Language Grammar item with English question stem [DATASET-D16] inconsistent with the stated Arabic-only design [Q30].

ID	Evidence
Q2	"Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region." (p.1)
Q30	"When designing ArabicMMLU, we excluded questions in English and only included questions in Arabic." (p.3)
Q37	"During manual data scraping, workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information (e.g., images, videos, or tables)." (p.4)
Q38	"If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)
Q92	"The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectal Arabic." (p.9)
D16	In the following Quranic verse, what is the correct parsing of the word كَلْبٌ — Grammar question with English question stem — unexpected language mixing in nominally Arabic benchmark

Strengths

- Text-only MSA format directly aligns with deployment's text-based modality [Q2, Q93, A4]
- Multimodal content systematically excluded; contextual passages preserved for dependent questions [Q37, Q38]
- Prompt language variants (Arabic/English) tested, providing some signal on cross-lingual prompt sensitivity [Q47]
- Uniform MSA register confirmed across all sampled configs [DATASET-D29, DATASET-D15]

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Q38	"If a question had additional contextual information (e.g., a passage referenced by several questions), the context was required to be included with each corresponding question." (p.4)
Q47	"We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)

Gap 1: Benchmark MSA register does not represent learner-register, code-switched Arabic produced by non-native users [Q92, elicitation expected_arabic_proficiency]

Recommendation: Construct a small bespoke evaluation set of learner-register and code-switched Arabic queries paraphrasing ArabicMMLU items, and measure performance degradation; this addresses the documentation gap noted in [WEB-6] where no learner-register Arabic benchmark exists.

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WEB-6	AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation (https://arxiv.org/abs/2510.13430)